

AFRICA ENERGY FINANCE SERIES · VOLUME 7

Bankable Hydro

*Structuring hydropower projects without creating
unsustainable public liabilities*

First edition, October 2026

Bankable Hydro. Structuring hydropower projects without creating unsustainable public liabilities.

Africa Energy Finance Series, Volume 7. First edition, October 2026.

This book is accompanied by the Bankable Hydro Integrated Bankability Model (Excel workbook, 35 sheets), a user manual, a course workbook and a case library. All references to "the model" are to that workbook.

The reference project used throughout, the Lumora Falls Hydropower Project in the Republic of Navaria, is fictional. Its river, utility, government, prices and contracts were constructed for teaching. Any resemblance to a real project is coincidental. Real projects appear only as benchmarks, drawn from public documents that are listed in the references; where a public figure could not be confirmed, the text says so.

Nothing in this book is investment, legal, tax or accounting advice. Readers who apply the methods to a real transaction should rely on their own advisers and on the project's own documents.

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PREFACE

Preface

Most of the hydropower projects that stall in Africa do not stall because the engineering is wrong. They stall because somebody further down the chain cannot carry the risk that the engineering leaves behind. The river is real, the dam can be built, the turbines will turn. Then the questions start. Who will buy the power, and with what money? Will the line to the load centre be finished on time, and who pays if it is not? What happens to a dollar tariff when the local currency loses a third of its value in a year? And when the lenders insist on guarantees, who in the Ministry of Finance has added them up?

This book is about those questions. It treats a hydropower project as one link in a chain that runs from the river to the national budget, and it asks whether the whole chain holds. The test we propose is simple to state and hard to pass: a project is ready for financial close only when it can deliver power that lenders will finance *and* when the commitments the state makes to get it financed are affordable under plausible stress.

We wrote it for people who sit on different sides of the same table. Developers and sponsors who want to know what lenders will ask before they ask it. Lenders and development finance institutions who need to see where a payment shortfall would actually come from. Ministry of Finance and PPP unit staff who are asked to sign guarantees and implementation agreements, often with little time and less analysis. Utility managers who will pay the bill for thirty years. Transaction advisers who have to reconcile all of them.

Three choices shaped the book. First, every claim about a real project is tied to a public source, and where we could not find one we say "public data not found" rather than estimate. Second, every number about the reference project comes from the companion model, so the text and the spreadsheet cannot drift apart. Third, we reviewed our own work adversarially before publishing it, and Chapter 18 reports what that review found, including the weaknesses we fixed and the ones we could not.

A word on tone. Hydropower finance attracts strong views. Large dams have their defenders and their critics, and both camps can point to evidence. This book takes no side on whether a given country should build a given dam. It takes a narrower position: if a country decides to build one, the deal should be structured so that its costs and risks are visible to the people who will eventually bear them.

ABBREVIATIONS

Abbreviations

Abbreviation	Meaning
CFADS	Cash flow available for debt service
COD	Commercial operation date
CV	Coefficient of variation
DFI	Development finance institution
DSA	Debt sustainability analysis
DSCR	Debt service coverage ratio
DSRA	Debt service reserve account
E&S	Environmental and social
EPC	Engineering, procurement and construction contract
FX	Foreign exchange
IDC	Interest during construction
IPP	Independent power producer
LCOE	Levelised cost of electricity
LIC-DSF	IMF and World Bank Debt Sustainability Framework for Low-Income Countries
LLCR, PLCR	Loan life and project life coverage ratios
MIGA	Multilateral Investment Guarantee Agency
NHSFO	Non-honoring of sovereign financial obligations (MIGA cover)
PFRAM	IMF and World Bank PPP Fiscal Risk Assessment Model
PPA	Power purchase agreement
PPP	Public-private partnership
PRG	Partial risk guarantee
P50, P90	Energy exceeded with 50% and 90% probability
QFD	Quasi-fiscal deficit
SOE	State-owned enterprise
VGF	Viability gap funding

PART I

The problem and the method

CHAPTER 1

Feasible, bankable, sustainable

Three ways a hydro project fails

A hydropower project can fail at three different points, and the three are easy to confuse.

The first is the familiar one. The scheme is technically feasible, the feasibility study is solid, and still no lender will commit. Usually the reason sits outside the plant. The buyer of the power has a record of paying late, the evacuation line has no financing, or the contract framework cannot be enforced in a way that a commercial bank's credit committee would accept.

The second failure is quieter. Lenders do commit, but only after the risk they would not carry has been moved somewhere else, almost always onto the state. A payment guarantee covers the utility. A loan guarantee covers the lenders. A termination clause promises that if the government defaults or a political event ends the concession, the state will repay the debt and compensate the equity. Each of these instruments makes the project bankable. None of them shows up as public debt on the day it is signed.

The third failure comes years later, when the commitments start to bite. The utility falls behind on its payments to the IPP. A guarantee is called. Energy that cannot be evacuated is paid for anyway under a deemed-energy clause. Arrears on a sovereign-guaranteed construction loan pile up. What was a power-sector problem becomes a fiscal one.

The public record contains all three. Zambia's Kafue Gorge Lower plant saw its cost rise from about USD 1.48 billion to about USD 2 billion [1], and the final figure includes roughly USD 312 million of capitalised interest [2]. By the time of the 2024 joint IMF and World Bank debt sustainability analysis, payables owed by the national utility ZESCO to independent producers, about USD 1.2 billion of domestic arrears, had been brought inside the sovereign debt perimeter [3]. Outside Africa, the World Bank reports that Lao PDR's public and publicly guaranteed debt reached 112 percent of GDP at the end of 2022, with the energy sector accounting for about 43 percent of it [4]. Much of that debt was built through take-or-pay contracts signed by a financially weak domestic utility.

These are not isolated cases. Across Sub-Saharan Africa, Trimble and co-authors estimated that the quasi-fiscal deficit of the power sector, the gap between what an efficient utility would need to recover its costs and what it actually collects, averaged about 1.5 percent of GDP and exceeded 5 percent in several countries [5]. A World Bank review of fiscal risk in the power sector found that annual fiscal injections into power SOEs averaged 0.25 percent of GDP and that between 24 and 41 percent of power PPP contracts were renegotiated [6].

The argument of this book

We argue that a hydropower project should be judged against three tests at once. The first is feasibility: can the plant be built and run to produce the energy claimed, at the cost claimed? The second is bankability: will commercial and development lenders finance it on terms the project can service, given the risks they are asked to carry? The third is fiscal sustainability, and it is the one most often skipped. Are the commitments the state makes to secure that financing affordable, including under the stresses that history tells us to expect?

Most analytical tools address one test at a time. Engineering studies answer the first. Project finance models answer the second, usually on the assumption that the offtaker pays. Fiscal risk tools such as the IMF and World Bank PFRAM answer the third at portfolio level, without a hydrological or utility layer [7]. Each tool is good at what it does. The difficulty is that the risks pass between them. A transmission delay that the project model ignores because the PPA pays deemed energy reappears as a utility cash shortfall, which reappears as a budget transfer or a guarantee call. If the three analyses are done by different teams on different assumptions, nobody sees the whole path.

What the book adds

We make four contributions.

The first is a single analytical engine, implemented in the companion model, that follows one set of assumptions from river flow to fiscal exposure. The same inputs drive the project's DSCR, the utility's cash position and the state's contingent liabilities.

The second is a small number of quantities that link the layers and that, as far as our review of existing products found, are not computed together elsewhere [8]. The most important are the offtaker's maximum sustainable PPA payment, the payment capacity gap that follows from it, and the transmission readiness gap. Chapters 6 and 8 define them.

The third is an explicit rule for where a shortfall goes. When the utility cannot pay, the money comes from a budget transfer, from a guarantee, or from nobody, in which case it becomes arrears owed to the project. The model routes every shortfall through that sequence, so a reader can see who ends up holding it in each scenario.

The fourth is the Hydropower Bankability Framework, a set of nine gates with published thresholds and a weakest-link rule. It is designed to be argued with. Every threshold is visible and editable, and Chapter 16 explains why we chose each one.

How to read the numbers

Every figure about the fictional Lumora Falls project comes from the companion model, run in the configuration stated in the text. The base case uses the PPP structure, a P50 generation case for cash flows and a P90 case for debt sizing. Stress cases are run with the

financing package locked at its base-case terms, because that is how lenders and sponsors test a package they have already signed. Where a figure says "about", it has been rounded from a model output, not estimated.

CHAPTER 2

A chain of risks

From river to budget

The chain this book follows has fifteen links:

hydrological resource, power plant, transmission, grid, demand, offtaker, utility, regulation, PPA, financing, PPP or IPP structure, government support, fiscal exposure, debt sustainability, financial close.

It is tempting to treat these as a checklist. They behave more like a pipe. Risk enters at one end, mostly as physical and commercial uncertainty about water, construction and demand, and it travels along the pipe until somebody absorbs it. Contracts determine where it stops. Very little of it disappears.

Risk is moved, not removed

Consider what a deemed-energy clause does. Suppose the transmission line is two years late. Without the clause, the project loses revenue and its lenders lose cover. With the clause, the utility pays for energy it cannot receive, so the project is whole and the lenders are content. The risk has not gone. It sits on the utility's income statement. If the utility cannot afford it, the government either tops up the utility or honours a payment guarantee. If the government does neither, the utility defaults on the PPA and the risk returns to the project, now in the form of arrears, which is worse than lost revenue because it comes with a dispute attached.

This is what we mean by saying risk is conserved. Contracts allocate it. Sometimes they allocate it to the party best able to manage it, which is the textbook ideal. Sometimes they allocate it to the party least able to refuse, which is closer to what happens when a single-buyer utility and a finance ministry face a well-advised sponsor with a deadline.

Form and substance

Off-balance-sheet structures make the point sharper. The Reventazón plant in Costa Rica is owned by a trust and leased to ICE, the state utility. In form, the asset and its debt sit outside ICE. In substance, construction, operating and overrun risk pass through to ICE under the lease, which is why rating analysts described the transaction as lightly structured ICE risk [9]. The structure did useful things. It allowed project-style financing, including an investment-grade bond and local-currency bank debt. It did not change who carried the risk.

We return to this in Chapter 12, where the model runs five financing structures for the same Lumora plant. Gross public exposure at commercial operation varies surprisingly little across them. What varies is the form the exposure takes and when it can be called.

The five layers

For analysis we group the chain into five layers, each with a question.

Table 2.1. The five analytical layers

Layer	Question	Main model sheets
Project	Does the plant produce enough, at a cost, to service its debt?	02 to 07, 14 to 21
Power system	Can the energy be evacuated and absorbed?	08 to 10
Offtaker and utility	Can the buyer pay, after everything else it must pay?	11, 12, 15, 16
Regulation and PPP	Is the contract chain enforceable and the risk allocation coherent?	13, 14, 17A, 29
Sovereign	What has the state promised, and can it afford it if called?	22 to 26

Notation

We use a small amount of notation. Time is indexed in years, t , from financial close. Operating years are indexed separately, $k = 1, 2, \dots$ from commercial operation. Money is in millions of nominal US dollars unless stated. E_t is energy in GWh, R_t revenue, O_t operating cost, T_t tax and DS_t senior debt service. The DSCR in year t is

$$\text{DSCR}_t = \text{CFADS}_t / \text{DS}_t, \text{ where } \text{CFADS}_t = R_t - O_t - T_t \quad (2.1)$$

Other symbols are defined where they are used.

PART II

The project layer

CHAPTER 3

Hydrology

Four kinds of capacity

Financing documents are full of capacity numbers, and they do not mean the same thing. It helps to keep four apart.

Installed capacity is the nameplate rating of the generating units. Available capacity is installed capacity multiplied by the share of time the units can run, after planned and forced outages. Expected generation is the energy the river allows on average, which depends on flow, head and how often the flow exceeds what the turbines can use. Contracted energy is whatever the PPA refers to, and it may differ both from what is delivered, because of curtailment, and from what is paid for, because of deemed-energy and take-or-pay clauses.

The gap between nameplate and energy can be very large. Belo Monte in Brazil has 11,233 MW installed against 4,571 MW of assured energy, about 41 percent of nameplate [10]. That is an extreme case, the result of a run-of-river design on a river with a pronounced dry season, but it shows why a lender looks past the MW figure.

From flow to energy

For each month, the model computes usable flow, average power and energy:

$$Q_u = \min(\max(Q - Q_e, 0), Q_d) \quad (3.1)$$

$$P = \min(\rho g Q_u H \eta_t \eta_g / 10^6, P_{inst}) \quad (3.2)$$

$$E = P \times \text{hours} \times \text{availability} / 1000 \quad (3.3)$$

Here Q is mean river flow, Q_e the environmental flow release, Q_d the rated turbine flow, H the net head, η_t and η_g the turbine and generator efficiencies, and P_{inst} the installed capacity. Power is in MW and energy in GWh.

For Lumora, mean monthly flows range from 120 to 520 m³/s, the design flow is 450 m³/s and the net head is 105 m. The result is a long-term mean, or P50, of about 1,666 GWh a year and a capacity factor of 47.5%. That sits close to the weighted average of 48 percent that IRENA reports for large hydro commissioned in Africa between 2018 and 2024, inside a 5th to 95th percentile band of 33 to 81 percent [11].

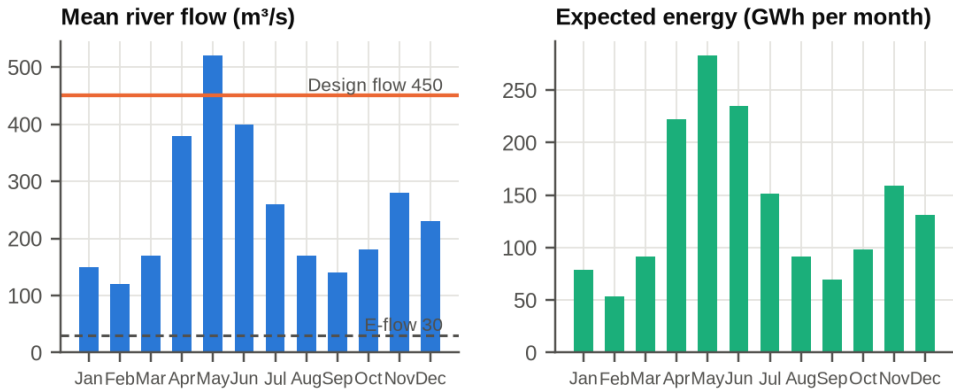


Figure 3.1. Lumora Falls: mean monthly flow and expected monthly energy. Flow above the 450 m³/s design flow is spilled; flow below the environmental release generates nothing. Source: companion model, sheet 03.

There is a bias hidden in equations 3.1 to 3.3 that matters for finance. Energy is a concave function of flow: it rises with flow until the turbines are full and then stops rising. Applying the function to *mean* monthly flows therefore overstates mean energy, because wet years cannot make up for dry years once the turbines are saturated. This is Jensen's inequality, and its size depends on how often flows cross the design flow. For a bankable estimate, the right procedure is to compute energy for each month of the historical record and average the results. The model flags this and accepts a simulated series if one is available.

P50, P75 and P90

Lenders size debt on a downside generation case. Practice varies, but the common reference is a P90 estimate, the annual energy exceeded in nine years out of ten. Without a full simulated series, the model uses a normal approximation driven by the coefficient of variation of annual energy:

$$P90 = P50 \times (1 - 1.2816 \times CV), \quad P75 = P50 \times (1 - 0.6745 \times CV) \quad (3.4)$$

With a CV of 12 percent, Lumora's P90 is about 1,410 GWh, a P90 to P50 ratio of 0.85.

Two cautions apply. The first is distributional. Annual hydro output is often skewed, with a longer tail on the dry side than the normal distribution allows, so the approximation can understate downside risk. The second is about horizon. A one-year P90 describes a single bad year. Debt service coverage over a tenor of fifteen or twenty years depends more on the average of many years, whose distribution is tighter, and on the risk of several dry years in a row, whose distribution is not captured by any single-year statistic. Applying a one-year P90 to every year of the loan life is conservative for the average and silent about persistence. We return to this in Chapter 18.

Drought and climate

Kariba is the reminder that persistence matters. In 2024 the Zambezi River Authority cut the water allocation for generation at Kariba from 30 to 16 billion cubic metres, according to news reporting at the time [12]. A drought of that kind does not produce one bad year followed by recovery. It produces a sequence.

Peer-reviewed work adds a regional dimension. Conway and co-authors found that planned hydropower expansion in eastern and southern Africa clusters capacity in basins with correlated rainfall, which raises the risk of concurrent climate-related supply disruption across interconnected systems [13]. A national power system that relies on several hydro plants in the same climate regime can lose a large share of its supply at once.

The model offers two hydrological stresses. A drought window applies a flow factor for a chosen number of consecutive operating years. The default, 55 percent of normal flow for three years, is set close to the cut in Kariba's 2024 allocation. A climate trend applies a percentage change in mean flow per decade, which we label illustrative until a basin-specific climate study supports a number. The IHA climate resilience guide sets out how such studies should be done [14].

With the financing package fixed at its base-case terms, a three-year drought at 55 percent of normal flow reduces Lumora's minimum DSCR from 1.64x to 1.44x. The equity return moves by less than a point, from 16.3% to 15.5%. The reason is the tariff. About 71% of Lumora's revenue comes from capacity payments, which do not depend on water. The project is protected. Chapter 10 asks who pays for that protection.

CHAPTER 4

Cost, construction and the reference class

Benchmarks and their limits

IRENA's cost database gives a weighted-average installed cost for large hydro in Africa of USD 2,330 per kW for projects commissioned between 2010 and 2017 and USD 2,515 per kW for 2018 to 2024, in 2024 dollars [11]. The World Bank appraisal of Nachtigal in Cameroon implies about USD 3,290 per kW, from a total of USD 1,383 million for 420 MW [15].

Unit costs in our case library spread much wider, from roughly USD 1,000 per kW for the Grand Ethiopian Renaissance Dam to more than USD 5,800 per kW for the 80 MW Rusumo Falls scheme. Those figures should not be compared directly. They are in the nominal currency of different years and they cover different scopes. Some include interest during construction, transmission and resettlement, and some do not.

Scope is the larger problem. The approved budget for Three Gorges ranged from 57 billion to 205 billion yuan depending on whether resettlement, grid connection and financing costs were counted [16]. A cost estimate that does not say what it includes cannot be checked. The model reports four cost measures separately: plant capital cost, the funding base (capital cost plus any project-built transmission), total uses (adding IDC, fees and the initial debt reserve), and system cost, which adds publicly funded transmission. For Lumora these are about 1,030 million real 2026 dollars of plant cost, 1,110 million nominal, and 1,316 million of total uses.

The reference class

The single most useful number in hydropower cost estimation comes from outside any one project. Ansar, Flyvbjerg, Budzier and Lunn studied 245 large dams built in 65 countries and found a median real cost overrun of 27 percent and a mean of 96 percent. Schedules overran by a median of 27 percent, or 1.7 years, and by 44 percent, or 2.3 years, on average. The average implementation time was 8.6 years [17].

The distribution is heavily skewed. The median tells you what to expect; the mean tells you how bad the tail is. African experience fits the pattern.

Table 4.1. Reported overruns and delays in selected African projects

Project	Reported overrun or delay	Source
Bui, Ghana	Cost raised by USD 168 million, about 27 percent, in a 2012 review	[18]

Project	Reported overrun or delay	Source
Rusumo Falls, regional	Main civil works contract up 47 percent	[19]
Karuma, Uganda	Commissioned in 2024 against a five-year contract signed in 2013	[20]
Kafue Gorge Lower, Zambia	Full commissioning in 2023 against a 2020 schedule	[1]
Isimba, Uganda	About seven months late; several hundred defects reported	[21]

Note: Figures as reported in the cited sources; scopes differ.

The model's default overrun stress is the median, 27 percent. That choice deserves comment. It means the stress case is roughly what an outside view would treat as an expected outcome, not a tail event. A user who thinks the project sits in the right-hand part of the distribution, because the geology is poorly investigated or the procurement is rushed, should run 50 percent or more.

Applied to Lumora with the financing package fixed, the 27 percent overrun opens a financing gap of USD 208 million, where the base case has none, and cuts the equity return to 9.3%. On the public side, the central-government fiscal NPV falls from about USD 49 million to USD –133 million, mainly because the public transmission line costs more, and peak contingent exposure rises to about USD 1,523 million. At the reference-class mean of 96 percent, the picture is of a different kind: the gap reaches USD 821 million, the equity return falls to 3.8% and contingent exposure peaks near USD 2,404 million. No sponsor would fund that gap without new public money, which is how overruns on this scale usually end.

Delay

Delay costs money twice. During construction, interest and escalation accumulate. After construction, revenue arrives later, so every dollar is worth less. At Kafue Gorge Lower, capitalised interest of USD 312 million made up about 16 percent of a USD 2 billion total [2]. The model adds a delay cost of 4 percent of capital cost per year of delay, extends the construction profile, and recomputes IDC. A two-year delay opens a financing gap of about USD 124 million and lowers the equity return to 11.1%.

CHAPTER 5

Operating cost, tariff and the cost of electricity

Operating costs

Hydropower is cheap to run. IRENA's sample suggests fixed operation and maintenance costs of 1 to 3 percent of installed cost a year, with an average a little under 2 percent and lower ratios for large plants [11]. Lumora's fixed costs, including insurance and a major maintenance reserve, come to roughly 2 percent. Two items that are easy to forget are the water-use royalty, which is a revenue stream for the state, and the community and environmental monitoring programmes that lenders' E&S standards require.

Plant cost and system cost

The model computes two levelised costs of electricity. The plant LCOE divides the present value of plant capital and operating costs by the present value of energy that is delivered or paid for as deemed energy. The system LCOE adds publicly funded transmission capital and operating costs and divides by energy delivered to load, net of transmission losses.

$$\text{LCOE}_{\text{plant}} = \text{PV}(C_{\text{plant}}) / \text{PV}(E_{\text{paid}}), \quad \text{LCOE}_{\text{system}} = \text{PV}(C_{\text{plant}} + C_{\text{tx,public}}) / \text{PV}(E_{\text{load}}) \quad (5.1)$$

At a 10 percent nominal discount rate, Lumora's plant LCOE is about USD 111 per MWh and its system LCOE about USD 134 per MWh. The difference is a real cost that sits on the public side of the ledger and that a plant-only analysis never sees. In negotiations between a sponsor and a ministry, it is common for the sponsor to quote the first number and for the ministry to pay the second.

A note on the discount rate. A nominal levelised cost at 10 percent is a convention, not a truth. IRENA's own cost of capital assumptions range from 3.8 percent in Europe to 12 percent in Africa, in real terms [11]. Readers comparing Lumora's LCOE with published figures should check the discount rate and whether the figure is real or nominal before drawing conclusions.

PART III

The power-system layer

CHAPTER 6

Transmission and the readiness gap

Can the plant evacuate what it generates?

The question sounds too obvious to ask. It is asked too rarely. The plant and its evacuation line are often procured by different entities, financed by different lenders and built to different timetables, and the interface between them is where a large share of value is lost.

Rusumo Falls illustrates the coordination problem. Generation was financed by IDA grants and credits to the three riparian states; transmission was financed in parallel by the African Development Bank and the European Union [19]. Supply into the grids depended on cross-border lines and three separate power purchase agreements. Uganda illustrates the financial consequence. In 2020, according to press reporting that we could see only in search results, about 9 percent of the transmission company UETCL's purchases were deemed energy, paid for but not evacuated [22].

Measuring readiness

We define the transmission readiness gap in two parts. The timing gap is the transmission line's commercial operation date minus the plant's. The capacity gap is installed capacity minus the evacuation capacity available in the plant's first operating year.

Evacuation capacity in each year is the smaller of what the line can carry and what the grid can absorb:

$$EVAC_t = \min(LINE_t , ABSORB_t) \quad (6.1)$$

where $LINE_t$ is the line capacity once the line is in service and the interim capacity of the existing network before that.

Grid absorption at minimum load is approximated as minimum load minus must-run generation plus export interconnector capacity. Delivered energy is then

$$E_t^{del} = \min(E_t \times \min(1, EVAC_t / P_{inst}) , ABSORBABLE DEMAND_t) \quad (6.2)$$

The proportional rule in equation 6.2 assumes curtailment scales with the shortfall in evacuation capacity. For a plant with daily pondage that can shift output into hours when the line has room, this is conservative. A dispatch study would do better.

Cost of the line

The Eastern Africa Power Pool master plan lists 400 kV AC line costs between about USD 0.36 and 0.70 million per km, in 2012 dollars and including IDC [23]. Lumora's 180 km line at USD 0.55 million per km, plus substations and downstream reinforcement,

costs about USD 219 million in real 2026 terms, roughly a fifth of plant cost and about USD 548 per kW of plant.

Who pays when the line is late

That depends on two contract choices: who builds the line, and whether the PPA pays deemed energy.

In the Lumora base case the state builds the line and the PPA pays deemed energy. With a two-year transmission delay and the financing package fixed, the IPP is untouched: its minimum DSCR stays at 1.65x, because the utility pays for energy it cannot receive. The utility's payment capacity ratio falls from 2.11x to 1.55x, and Gate 4 drops to critical.

Where the cost lands depends on how wide one draws the public sector. Central-government fiscal NPV hardly moves, from USD 49 million to USD 47 million. The consolidated figure, which includes the state utility, falls from USD –109 million to USD –227 million, because the utility pays deemed energy while still buying replacement power. An earlier version of our model, which stopped at the Treasury, showed the delay as fiscally beneficial. Chapter 18 explains the correction. The general point stands: a fiscal indicator that excludes the state utility will miss exactly the costs that power-sector contracts are designed to push onto it.

CHAPTER 7

Grid and demand

Four kinds of demand

A load forecast can mean four different things, and only one of them supports a loan.

Table 7.1. Demand concepts used in the model

Concept	Definition
Potential	Sum of segment demand (residential, commercial, industrial, mining, productive use, public) plus export potential
Commercial	Segment demand weighted by the share coming from customers able and willing to pay cost-reflective tariffs
Contracted	Energy covered by the PPA, at P50
Bankable	The lesser of contracted energy and absorbable energy scaled by the commercial share

Absorbable energy is the supply gap left after other committed plants, plus existing thermal generation the project can displace, plus signed export contracts. The distinction between potential and commercial demand is where electrification ambitions meet balance sheets. A household that would use electricity if it were connected is potential demand. A household that is connected and pays its bill is commercial demand.

Building ahead of demand

Tanzania shows the risk of building ahead of demand. With Julius Nyerere's 2,115 MW in service, national installed capacity reached about 4.6 GW against a peak demand of about 2.27 GW, a surplus of 2.38 GW, and exports were being discussed to absorb it [24]. A surplus of that size is not a failure if demand grows into it. It is a fiscal question in the meantime, because somebody pays for capacity that is not used.

Grid integration

A plant that supplies a large share of system peak raises questions about reserves, frequency control and what happens when it trips. Lumora at commercial operation is about 19% of national peak. The model treats a share above 15 percent as conditional and above 25 percent as a development gap. These are screening thresholds, not engineering limits, and a grid-impact study should replace them.

PART IV

The offtaker and the contract

CHAPTER 8

The utility and the maximum sustainable PPA payment

The assumption most models make

A standard project finance model has a revenue line, and the revenue line assumes the buyer pays. Sensitivity tables may haircut generation or tariff, but the counterparty's ability to pay is treated as a credit question for someone else. In Sub-Saharan Africa that is a dangerous habit.

The World Bank's work on utility finances gives the reasons. Transmission and distribution losses across the region averaged about 15 percent of energy dispatched, and 23 percent excluding South Africa, against a benchmark of about 10 percent. Bill collection ranged from about 58 percent in the weakest utilities to close to 100 percent in the best [5]. In Kojima and Trimble's 39-country assessment, only two power sectors were found to be financially viable and about half covered their operating costs [25]. (We saw these two figures in a summary rather than the full report, and they should be checked against it.) The case files repeat the theme at project level. Ghana's ECG, for example, owed the Bui Power Authority about USD 612 million in March 2023 [26].

A cash model of the oftaker

The companion model builds a simplified cash model of the utility. All local-currency flows are translated into US dollars at the projected exchange rate, so that the utility's ability to pay can be compared directly with a dollar PPA bill.

$$\text{CASH}_t^{\text{pre}} = \text{COLL}_t + \text{TR}_t - \text{OPEX}_t - \text{OS}_t - \text{EDS}_t \quad (8.1)$$

Collections *COLL* are sales times the retail tariff divided by the exchange rate, times the collection rate. *TR* is government transfers, *OPEX* the utility's own operating costs, *OS* the cost of supply from sources other than the new plant, and *EDS* debt service on existing borrowing. Sales are limited by the energy actually available: other supply is capped at what existing and planned plants can deliver, and demand above that is unserved. When Lumora comes online it either serves demand that was going unmet or displaces other supply, and the cash model credits the utility with whichever applies.

From this we define the maximum sustainable PPA payment:

$$\text{MSP}_t = \max(0, \text{CASH}_t^{\text{pre}}) / c \quad (8.2)$$

where *c* is a coverage buffer, 1.2 by default. The buffer represents the share of free cash a prudent utility, or its regulator, would not commit to a single supplier. The oftaker payment capacity gap is then

$$GAP_t = \max(0, PPA_t - MSP_t) \quad (8.3)$$

where PPA_t is the utility's share of the project's billed revenue.

Where the gap goes

The model sends the gap through three doors in a fixed order. If the government has committed to backstop the utility from the budget, the gap becomes a fiscal transfer. If not, and the structure includes a sovereign payment guarantee, the gap becomes a guarantee call. If neither applies, the gap becomes arrears owed to the project, which reduce its cash revenue and therefore its debt cover.

$$BACKSTOP_t = b \times GAP_t, \quad CALL_t = \min(GAP_t - BACKSTOP_t, L_t - OUT_{t-1} + REIMB_t) \quad (8.4)$$

$$UNPAID_t = GAP_t - BACKSTOP_t - CALL_t \quad (8.5)$$

Here b is the share of the gap the budget covers, L_t the guarantee limit (twelve months of billing by default), OUT the calls not yet reimbursed and $REIMB$ what the utility repays when its cash recovers. A guarantee is a liquidity instrument with a ceiling. Once it is exhausted, the shortfall becomes arrears owed to the project. The sequence is a modelling choice; in practice governments sometimes pay late rather than not at all. The point of the routing is to make the destination visible, not to predict the exact timing.

What breaks the utility

In the Lumora base case the utility can pay. Its payment capacity is about 2.1x times the PPA bill in the worst of the first ten operating years, and there is no gap.

Being able to pay is not the same as being better off. Lumora's blended tariff is about USD 109 per MWh in its second year. The utility collects less than that for each megawatt-hour it sends out once losses and non-payment are taken into account. Every unit it buys from Lumora therefore costs it money, even when the unit serves demand that would otherwise go unmet. Over the life of the PPA the effect is a present-value loss to the utility that turns a small positive central-government fiscal NPV of about USD 49 million into a consolidated figure of about USD -109 million. Neither the project's lenders nor a Treasury-only fiscal analysis would see it.

Two stresses change that.

The first is an offtaker stress. From commercial operation, collections fall by 8 percentage points, government transfers are cut by 30 percent and the retail tariff is frozen in nominal terms for five years before indexation resumes. Payment capacity falls to zero within the first ten years. Over the life of the PPA the cumulative gap is about USD 5,460 million in nominal terms. With the budget backstop in place, central-government fiscal NPV moves from about USD 49 million to about USD -1,177 million, and the project's lenders see nothing.

Remove the backstop and the picture changes completely. The twelve-month payment guarantee is exhausted in the first year of the shortfall, the remainder becomes arrears, and the project cannot service its debt: minimum DSCR falls to $-0.69x$ and the sponsors lose their equity. The state does not escape. Its consolidated fiscal NPV is about USD $-1,180$ million, because the arrears sit on the utility's books. The choice is not between paying and not paying. It is between paying through the budget and paying through a default.

The second is a one-off devaluation of 50 percent at commercial operation. The PPA is in dollars, retail tariffs are in local currency and indexation does not catch up with the step. Payment capacity falls to $0.57x$ of the bill, the lifetime gap is about USD 405 million, and the consolidated fiscal NPV falls to about USD -496 million.

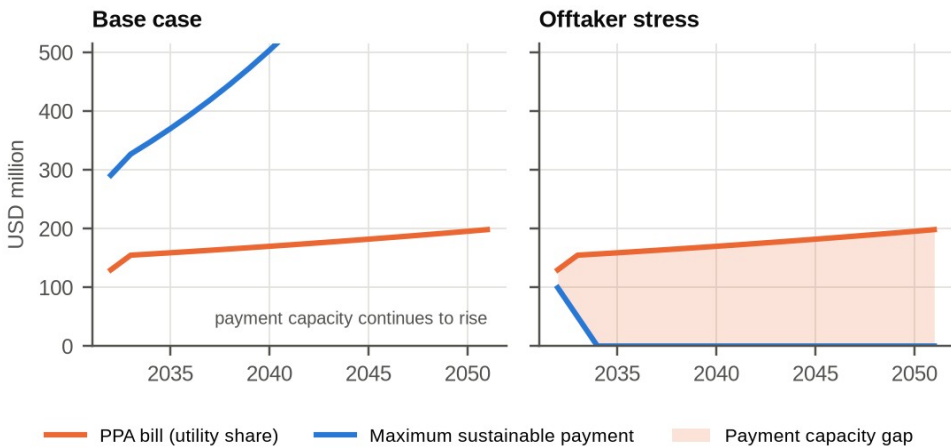


Figure 8.1. Utility PPA bill against maximum sustainable payment, base case and off-taker stress (USD million, nominal). The shaded area is the off-taker payment capacity gap. Source: companion model, full-engine runs.

The numbers are large because the mechanism compounds. In the fictional base case, the utility's cash margin is a modest share of its collections. A permanent price shock, whether a tariff freeze or a devaluation without pass-through, lowers that margin every year for thirty years. This is the mechanism by which a project that is bankable on its own terms becomes a fiscal problem, and it is invisible in any model that assumes the off-taker pays.

Limits of the utility model

The utility model is deliberately simple. It has no balance sheet, it treats the other supply mix as half dollar-linked, and it lets operating costs scale with sales volume. A real assessment should start from audited accounts and a tariff path agreed with the regulator. The IMF's SOE health-check and stress-test tools provide a fuller framework for that

work [27, 28]. What the simple model does well is make the question unavoidable: before negotiating the tariff, compute what the buyer can pay.

CHAPTER 9

Regulation as an input to bankability

A readiness screen, not a legal opinion

The model's regulatory matrix scores sixteen items as ready, partial, gap or critical gap. The items are the electricity law, the generation licence, water rights, the concession or implementation agreement, PPP legislation, the IPP procurement framework, enforceability of the PPA, tariff regulation, regulator independence, grid access, the grid code, environmental and social permits, land rights, foreign exchange convertibility, the tax regime and the rules on repatriation.

The matrix is a development screen. It asks whether each element is in a state that lenders' counsel would accept, not whether it is good law. Gate 6 counts the items at gap and critical gap and converts the counts into a status.

Governance decides who lends

The case files suggest that governance and procurement failures stop more projects than technical ones. At Gibe III in Ethiopia, the main civil works contract was awarded without competitive tender, which was incompatible with World Bank procurement policy; the African Development Bank and the European Investment Bank withdrew in 2010, and the electro-mechanical package was financed by ICBC [29, 30]. The first attempt at Bujagali in Uganda collapsed after the export credit agencies withdrew in January 2002, a month after Board approval, and corruption investigations involving an EPC contractor followed [31]. At Batoka Gorge, Zambia withdrew from the 2019 arrangement in June 2023, citing improper procurement, and the project is being re-tendered [32].

We draw a modest conclusion from this. Transparent, competitive procurement is part of bankability. It decides which lenders can participate, which in turn decides tenor, pricing and the guarantees the state is asked to provide.

Environment and society

Gate 9 covers environmental and social readiness: compliance of the ESIA and management plans with lender standards such as the IFC Performance Standards or the Hydropower Sustainability Standard [33], the status of resettlement, and transboundary issues. The Grand Ethiopian Renaissance Dam shows that transboundary questions can outlast construction; at the time of our research there was still no agreement with Egypt and Sudan on filling and operation [34]. Lenders bound by safeguard policies treat these questions as conditions, not as afterthoughts.

CHAPTER 10

The PPA: who carries water, volume, currency and termination risk

The two-part tariff

Most African hydro IPPs use some version of a two-part tariff. A capacity or availability charge, in dollars per kW per month, pays for fixed costs, debt service included, as long as the plant is available. An energy charge, in dollars per MWh, pays for energy delivered. The larger the capacity share, the less hydrology risk the IPP carries and the more the buyer carries.

Lumora's capacity charge is USD 26 per kW per month and its energy charge USD 25 per MWh, in 2026 dollars, which puts about 71% of lifetime revenue on the capacity side. That is why the drought stress in Chapter 3 barely touched the equity return. The protection is not free. In a drought the utility pays capacity charges for a plant that produces less, and it buys replacement energy elsewhere, often from thermal plants at a higher cost.

Volume clauses

Deemed-energy clauses pay for energy that the seller could have delivered but the buyer or the grid could not take. Take-or-pay clauses set a floor on annual payments regardless of offtake. Both convert system risk into utility risk, and both are reasonable when the utility controls the cause. They become problematic when the utility controls neither the transmission timetable nor demand growth, because then the clause allocates risk to the party that can do least about it.

Currency

A dollar tariff moves currency risk from the project to a buyer whose revenue is in local currency. The model allows part of the tariff to be denominated in local currency and indexed to local inflation, which shares the risk. Local-currency debt is the other half of the solution. Nachtigal raised a 21-year local-currency tranche guaranteed by IBRD alongside an 18-year euro tranche [15], and Reventazón paired dollar development-bank and bond debt with local-currency bank loans [9]. Neither eliminates the problem, but both reduce the mismatch at its source.

Termination

Termination provisions are where the largest contingent liabilities hide. A typical formula for termination following government default or political force majeure pays outstanding senior debt plus some compensation to equity. In the model, exposure in each year equals debt outstanding plus unrecovered private equity times one plus a premium, 15 percent by default. For Lumora under the PPP structure, this is roughly USD 960 million at commercial operation in the closed-form comparison. Few ministries record this number anywhere.

PART V

The finance layer

CHAPTER 11

Sizing hydro debt

Tenor

For an asset that can run for fifty to a hundred years, the length of the debt matters more for the tariff than almost any other financing term. Commercial tenors for African hydro are rarely longer than fifteen to eighteen years, as Ebobisse and Hott noted [35]. Nachtigal's tranches ran eighteen and twenty-one years with a grace period of about six years [15].

Bujagali shows the cost of getting it wrong. Its original debt tenor was short, so capacity charges were front-loaded and the tariff was high in the early years. The 2018 refinancing extended maturity and lowered the tariff, but the state paid part of the price: the 30 percent corporation tax was waived in exchange for the re-profiling, and the return on equity was cut from 19 to a proposed 15 percent [36, 37].

Sculpting without circularity

The model sizes commercial debt by sculpting debt service to a target coverage ratio on a lender case. In each loan year k ,

$$DS^m_k = \max(0, CFADS^L_k / DSCR^* - DS^c_k) \quad (11.1)$$

where $CFADS^L$ is lender-case cash flow (P90 generation), $DSCR^*$ the target ratio and DS^c concessional debt service, which is fixed by the concessional loan's own terms. Debt capacity is the present value of that stream at the commercial interest rate r^* :

$$D^m_{\max} = \sum_k DS^m_k / (1 + r)^k \quad (11.2)$$

and commercial debt is the lesser of capacity and the gearing limit less concessional debt.

A model in which debt funds its own interest during construction is circular, because IDC depends on debt and debt depends on IDC. Practitioners often break the loop with a macro or with Excel's iterative calculation. We avoid it with a closed form. If debt is drawn in proportion to capital spending, IDC per dollar of debt is a constant k that depends only on the drawdown profile and the interest rate. With a gearing limit g , upfront fees f , capital cost K and concessional debt D^c with its own factor k^c , the commercial debt that respects the limit is

$$D^m = [g(K + D^c(k^c + f)) - D^c] / [1 - g(k^m + f)] \quad (11.3)$$

The lender case computes tax on depreciation that excludes IDC and fees, which is slightly conservative. The initial debt service reserve is funded by equity.

For stress testing, the user can lock the financing package. In locked mode every tranche amount, the grant and the government's equity stay at their base-case values, the commercial loan keeps its base-case principal schedule, and interest floats only on the share that is not hedged. Equity absorbs the rest. That is the position a sponsor is in after financial close, and it is how lenders test a package they have already sized.

Covenants

We found little institutional evidence on covenant levels for African hydro. The most specific public source is Nachtigal's appraisal document, which reports a lenders' base-case minimum DSCR of 1.45 and an average of 1.53, described as consistent with precedent [15]. The model's sizing ratios, between 1.20 and 1.35 depending on structure, and its distribution lock-up at 1.10 are therefore user inputs, not sourced norms.

Lumora base results

Table 11.1. Lumora Falls, PPP structure, base case (debt sized in the model)

Metric	Value
Plant capital cost, nominal	USD 1,110 million
Total uses	USD 1,316 million
Senior debt (concessional plus commercial)	USD 960 million
Senior debt capacity	USD 1,108 million
Minimum and average DSCR	1.64x and 3.32x
LLCR at commercial operation	1.87x
Project IRR, post-tax, nominal	9.7%
Private equity IRR, nominal	16.3% against a 15.0% target
Financing gap	USD 0 million

Note: Financing gap is private equity required beyond the amount sponsors are assumed willing to commit (20 percent of the funding base). IDC and fees are debt-funded within the 75 percent gearing limit.

Two things stand out. The minimum DSCR in the P50 case is well above the sizing target, because debt is sized on P90 and, in this structure, limited by gearing rather than by cover. And the equity return clears its target with no financing gap. On its own terms Lumora is bankable. Whether it should be built on these terms is a separate question, which Chapters 12 to 14 take up.

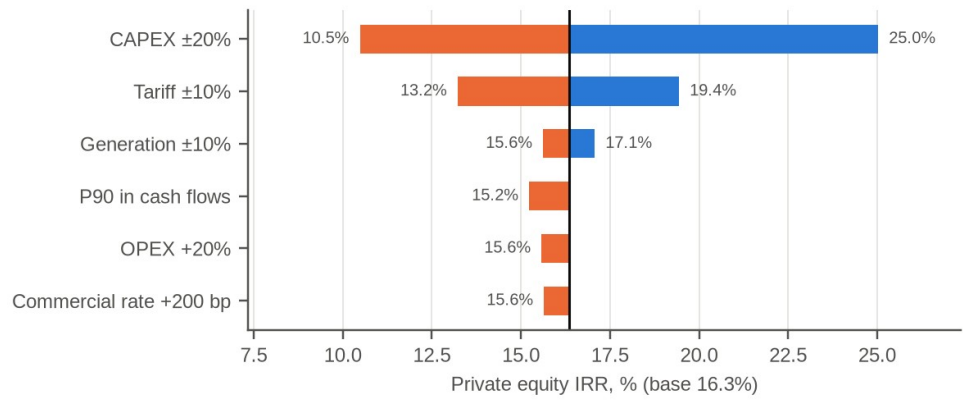


Figure 11.1. Sensitivity of the private equity IRR to single-variable changes, Lumora PPP structure, debt fixed at base-case amount. Source: companion model, full-engine runs.

CHAPTER 12

Five structures, one project

The structures

The model holds five financing structures for the same plant.

1. A public structure, in which a state-owned company builds and owns the plant with debt borrowed by the sovereign.
2. An IPP, in which a private sponsor builds, owns and operates the plant under a thirty-year concession.
3. A PPP in which a state-owned company and a private sponsor share the equity and development finance institutions provide concessional debt.
4. A hybrid, in which the state funds the civil works and a private company finances the electro-mechanical equipment and operates the plant.
5. A blended structure, with grants, concessional debt and private capital in proportions typical of climate-finance packages.

Each structure has its own debt terms, equity mix, guarantee package and target return.

A closed-form comparison

Before running the full engine, it is useful to ask what tariff each structure would need to recover its costs. Treating the funding as a level annuity over the thirty-year concession, the required tariff for structure s is

$$\tau_s = [K (1 - \gamma_s) \times CRF(w_s, N) + O] / E_{P50} \quad (12.1)$$

where K is the funding base grossed up for debt-funded IDC and fees, γ_s the grant share, w_s the pre-tax cost of capital on non-grant funding, CRF the capital recovery factor over N years, and O annual operating cost. Private equity returns are grossed up for corporate tax. Government equity is costed at the government's discount rate, not at zero, because public money has an opportunity cost. The formula ignores tenor mismatch, the debt reserve and the tax holiday, so we also solve the full engine for the tariff each structure needs to reach its target equity return.

Table 12.1. Closed-form comparison of the five structures (Lumora P50, nominal)

Metric	Public	IPP	PPP	Hybrid	Blended
Cost of non-grant capital	4.5%	12.8%	7.8%	8.1%	7.8%
Required tariff, USD/MWh	66	118	82	61	79
Upfront public capital, USD m	167	0	167	389	144
Direct public debt, USD m	944	0	0	0	0

Metric	Public	IPP	PPP	Hybrid	Blended
Guaranteed debt, USD m	0	0	333	122	120
Termination exposure, USD m	0	1,160	960	738	991
Gross public exposure at COD, USD m	1,110	1,160	1,127	1,127	1,135
Gross public exposure, % of GDP	3.2%	3.3%	3.2%	3.2%	3.2%

Note: Gross exposure equals public capital plus direct debt plus the larger of (guaranteed debt plus twelve months of PPA payments) and the termination amount. Screening calculation; see Chapter 13 for the full-engine measures.

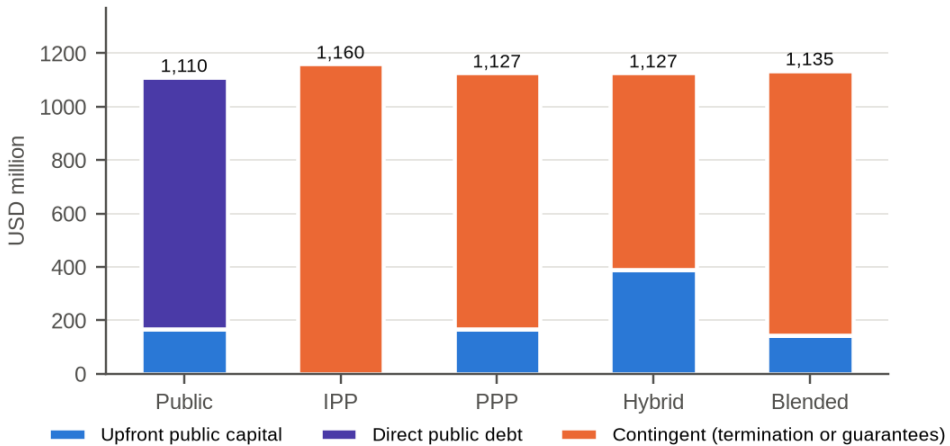


Figure 12.1. Composition of gross public exposure at commercial operation by structure (USD million). The total barely changes; its form does. Source: companion model, sheet 17A.

What the comparison shows

The required tariff depends mostly on the cost of capital. In the screen the IPP needs about USD 118 per MWh and the public structure about USD 66. The full engine, which includes tenors, reserves and the tax holiday, puts the IPP's requirement higher: about USD 137 per MWh in the second operating year to reach its 16 percent target, against about USD 105 for the PPP, USD 94 for the blended structure and USD 75 for the hybrid. The difference between the IPP and the rest is the price of private risk-taking. Whether it is worth paying depends on whether the private sponsor actually takes risks the state would otherwise carry, such as construction and operating performance.

Gross public exposure at commercial operation lies between about USD 1,110 million and USD 1,160 million, roughly 3.2% of GDP, in every structure. This is close to an identity rather than a finding. Every dollar of capital is either public money or protected by a termination payment, so the total cannot stray far from the funding base. What the structures change is the form and timing of the exposure. The public structure carries it as direct debt. The IPP carries it as a termination obligation and a payment guarantee. The PPP and the blended structure carry it as a mix of public capital,

guaranteed debt and termination exposure. Form matters, because direct debt is recorded and a termination obligation usually is not, and because a guarantee can be called in a year when the state is least able to pay.

The full engine

The closed form holds the tariff at cost recovery. The full engine holds it at the negotiated PPA level, which is the same for every structure in our runs. That difference produces two results that matter for policy.

First, grants without a tariff adjustment are a windfall. In the hybrid structure, the state contributes 35 percent of the funding base as a capital grant. At the unchanged PPA tariff, the private equity return rises to 30.3% while the fiscal NPV falls to about USD –297 million. A grant or viability gap payment should be matched by a lower tariff or an explicit cap on returns. Otherwise public money flows to private shareholders.

Second, public ownership at an IPP-level tariff looks attractive to the treasury and expensive to consumers. In the public structure, fiscal NPV is about USD 411 million at the same tariff, because the state receives all equity returns. Consumers, meanwhile, pay a blended tariff of about USD 109 per MWh in the second operating year, well above the public structure's cost-recovery level of about USD 66.

Table 12.2. Full-engine results by structure (base case, debt sized in the model)

Structure	Equity IRR	Min DSCR	Financing gap, USD m	Fiscal NPV, USD m	Consolidated NPV, USD m	Peak contingent exposure, USD m
Public	n/a	1.74x	0	411	252	0
IPP	10.4%	1.41x	87	–41	–200	1,429
PPP	16.3%	1.64x	0	49	–109	1,221
Hybrid	30.3%	2.36x	0	–297	–456	1,238
Blended	18.5%	1.89x	0	–12	–170	1,320

Note: Public structure has no private equity, so no equity IRR is reported. Its exposure appears as direct debt, not as a contingent liability.

No structure is free. The useful question is not "PPP or public?" but which combination of tariff, public capital and risk transfer minimises the total cost to consumers and taxpayers, within an exposure the state can afford.

PART VI

The public-finance layer

CHAPTER 13

Direct support and contingent liabilities

Two kinds of public commitment

The state supports a hydro project in two ways. Direct support is cash it actually spends: equity in a project company or state utility, capital grants and viability gap payments, the cost of building the transmission line, and budget transfers to the utility. Contingent liabilities are payments it may have to make if something happens: a guaranteed loan defaults, the utility misses PPA payments covered by a sovereign guarantee, the concession is terminated, or the central bank cannot provide foreign exchange.

The literature on this distinction is long. Polackova's 1998 paper is still a good introduction to the idea that contingent liabilities are a hidden form of fiscal risk [38]. More recent work has measured what happens when they materialise. Using a dataset of realised contingent liabilities, Bova and co-authors estimated an average fiscal cost of about 6 percent of GDP per episode; the breakdown by type, about 1.2 percent for PPPs and 3 percent for SOEs, we saw only in a summary and could not confirm in the full text [39].

Multilateral guarantees are sovereign exposure

It is common to hear that a World Bank or African Development Bank guarantee "de-risks" a project. It does, for the lenders. For the state, a guarantee from a multilateral bank to commercial lenders is backed by an indemnity from the government to the multilateral bank. Uganda signed an indemnity agreement with IDA on 18 July 2007 to support the Bujagali partial risk guarantee of up to USD 115 million [40]. Cameroon backs the IBRD payment guarantee of up to EUR 86 million and loan guarantee of up to EUR 171 million for Nachtigal through a similar indemnity [41]. The guarantee changes the lender's counterparty from the utility or the state to a triple-A institution. It leaves the state's obligation in place.

That is not an argument against such guarantees. They lengthen tenor, lower pricing and bring in lenders who would otherwise stay away, and those benefits can far exceed the cost. It is an argument for counting them.

Three measures, kept apart

The model reports three measures of contingent exposure and does not add them together.

The first is the maximum simultaneous exposure in each year:

$$X_t^{\max} = \max(\text{TERM}_t, \text{DG}_t + \text{PG}_t + \text{FXG}_t) \quad (13.1)$$

where *TERM* is the termination amount, *DG* the guaranteed share of senior debt, *PG* twelve months of PPA payments under a sovereign payment guarantee and *FXG* foreign exchange cover on a year's remittances. Termination and the guarantees are not added because termination would replace them: once the state has bought out the project, it no longer guarantees its debt, its PPA or its remittances. The termination amount is outstanding debt plus the larger of unrecovered equity with a 15 percent premium and the present value, at the sponsor's target return, of the distributions equity would still have received.

The second is the set of calls the model actually computes in the active scenario: budget backstop payments, guarantee calls and minimum-revenue payments. These are deterministic. In the offtaker stress, they are the mechanism by which the utility's gap becomes a fiscal cost.

The third is an expected loss, computed as user-entered annual call probabilities times exposure times loss given call. We include it because ministries are often asked for a single number, and we label it clearly as a judgement. The probabilities in the model are not calibrated default rates, and the expected loss is only as good as they are. A number presented as an expected value acquires an authority it has not earned. We would rather show the maximum exposure and the scenario calls and let the reader judge the likelihood.

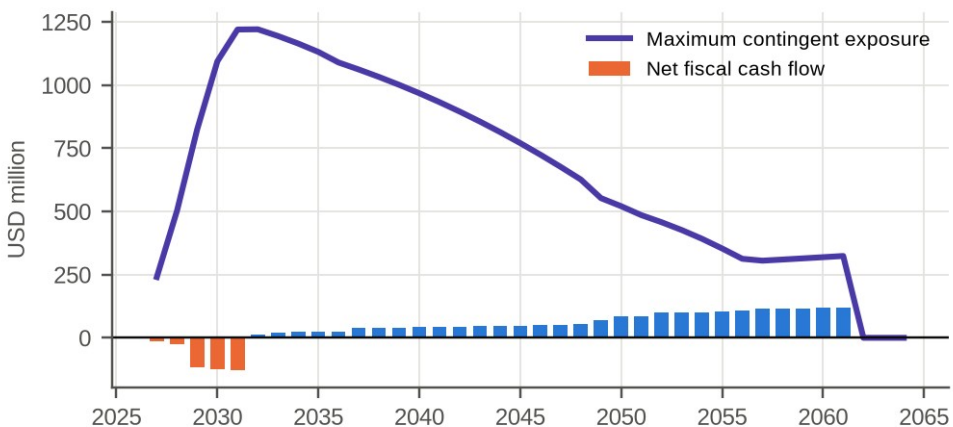


Figure 13.1. Net fiscal cash flow to the government (bars) and maximum contingent exposure (line), Lumora PPP base case, USD million nominal. Source: companion model.

Lumora's fiscal profile

In the base case the state commits government equity, a 5 percent viability gap contribution and the full cost of the transmission line, and receives tax, royalties and dividends in return. Peak annual cash need is about 1.8% of government revenue, in the

construction years. Peak contingent exposure is about USD 1,221 million, or 2.7% of GDP, reached around commercial operation and declining as debt is repaid.

The fiscal NPV depends on where one draws the boundary of the state. For central government alone, at an 8 percent discount rate, it is about USD 49 million, positive but small. Including the state utility, it is about USD –109 million, because the utility pays more per megawatt-hour than it recovers. The consolidated figure is the one a finance ministry should use, since the utility's losses come back to the budget as transfers, arrears or recapitalisation sooner or later [6].

The stresses show how thin the margin is. The offtaker stress and the currency step turn both measures deeply negative, as Chapter 8 showed, and they do so without any change in the project's own debt cover.

CHAPTER 14

Project-level fiscal exposure screening

What a screen can and cannot do

The IMF and World Bank Debt Sustainability Framework for Low-Income Countries sets thresholds for debt burden indicators according to a country's debt-carrying capacity. For the present value of public and publicly guaranteed external debt, the thresholds are 30, 40 and 55 percent of GDP for weak, medium and strong capacity; the corresponding benchmarks for total public debt are 35, 55 and 70 percent [42]. The framework also includes a contingent liability stress test, with a default shock of 2 percent of GDP for SOE debt and 35 percent of the PPP capital stock where that stock exceeds 3 percent of GDP [42]. PFRAM 2.0 is the standard tool for assessing the fiscal costs and risks of individual PPPs and portfolios [7, 43].

The screen in the companion model does not replace either. It answers a narrower question that ministries face in practice, often under time pressure: does this one project add material fiscal pressure, and should it be escalated to the debt management office before signature?

Four tests

The screen applies four tests:

1. Does the project's on-budget debt, plus cumulative fiscal outlays and any utility arrears to the project, add more than 1 percent of GDP in any year?
2. Does peak contingent exposure exceed 2 percent of GDP?
3. Does the peak annual cash requirement exceed 1 percent of government revenue?
4. Does the project push public debt above the user's benchmark when it was below it before?

The results are combined with the latest published DSA risk rating. The screen returns low pressure when no test is breached and the rating is low or moderate, high pressure when three or more tests are breached or when any test is breached in a country already at high risk or above its debt benchmark, and moderate pressure otherwise. All thresholds are editable; the defaults are screening conventions that we chose, not official limits.

Lumora

In the base case, Lumora breaches two tests: peak contingent exposure of about 2.7% of GDP and peak cash requirement of about 1.8% of revenue. With Navaria's fictional debt at 48 percent of GDP and a moderate DSA rating, the screen returns moderate pressure.

The cost overruns, the offtaker stress and the currency step move it to high. Table 17.1 lists the result for every scenario.

PART VII

Decision

CHAPTER 15

Who bears what

The allocation matrix

The model maps sixteen risks against eight parties: government, developer, EPC contractor, lender, utility, consumer, insurer and development finance institution. For each risk it records the primary bearer, the parties that share it, whether it can be mitigated by contract, the main instruments, and the model stress that quantifies it. The allocation is indicative of a typical PPP and should be replaced by the actual contract positions.

Table 15.1. Selected risks, typical allocation and instruments

Risk	Typical primary bearer	Main instruments	Model stress
Hydrology	Developer (energy charge)	Two-part tariff, P90 sizing, reserve accounts	Drought, climate
Construction and overrun	EPC and developer	Fixed-price date-certain EPC, liquidated damages, contingency	CAPEX overrun, delay
Demand and transmission	Utility	Deemed energy, take-or-pay, interface agreements	Low demand, transmission delay
Offtaker payment	Utility, then government	Letters of credit, liquidity facilities, sovereign guarantees	Offtaker stress
Currency	Utility and government	Dollar tariff, local-currency tranches, convertibility undertakings	FX step
Political	Government	MIGA cover, partial risk guarantees, arbitration	Termination exposure

Instruments and where they lead

The instruments available to mitigate payment and political risk are well known. MIGA offers cover against the non-honouring of sovereign financial obligations [44]. The World Bank offers loan and payment guarantees [45, 46]. The African Development Bank provides partial risk guarantees, as for the Sahofika project in Madagascar [47]. African Trade and Investment Development Insurance runs a regional liquidity support facility that, according to its published description, can cover up to twelve months of revenue [48].

Almost all of them move risk towards the state or towards a multilateral institution that the state indemnifies. A liquidity facility is a partial exception, because it bridges temporary shortfalls rather than absorbing permanent ones. That is why the matrix counts how many risks end with the government as primary bearer. In the Lumora PPP, it is the largest single bearer of risk, ahead of the developer.

CHAPTER 16

The Hydropower Bankability Framework

Design principles

We built the framework around four principles.

It should be explainable. Each gate is a short list of tests, each test is a named metric, and each metric has three visible thresholds that separate ready, conditional, development gap and critical gap.

It should be auditable. Every metric is a cell in the model, and every threshold is an input that a reviewer can see and change.

It should have no hidden weights. A gate takes the status of its weakest test, and the overall verdict follows from the weakest gate. Weighted scores look precise and hide trade-offs that should be argued over. A project with excellent hydrology and an insolvent offtaker is not "70 percent bankable".

It should be honest about its thresholds. We set them as screening values and say so. They should be calibrated to the lender term sheet, the country's fiscal rules and the project's context.

The gates

Table 16.1. The nine gates and their tests

Gate	Tests
1. Resource	Years of reliable flow record; P90 to P50 ratio; maturity of the hydrology study
2. Demand	Bankable to contracted demand in the worst of the first five years; project share of system peak
3. Configuration	Unit cost against benchmark; capacity factor; maturity of the feasibility study
4. Transmission	Evacuation capacity at commercial operation against installed capacity; transmission lag; transmission financing secured
5. Offtaker and utility	Payment capacity against PPA bill in the worst of the first ten years; collection rate; months of payment security
6. Regulation and PPA	Regulatory items at critical gap; items at gap; fixed share of revenue
7. Financing	Minimum DSCR; financing gap against total uses; equity return against target

Gate	Tests
8. Public finance	Peak cash need against revenue; peak contingent exposure against GDP; peak on-budget increment against GDP; consolidated fiscal NPV against GDP; DSA rating
9. Environment and society	ESIA compliance; resettlement status; transboundary issues

A test scores 3 when it meets the ready threshold, 2 when it meets the conditional threshold, 1 when it meets the development-gap threshold and 0 otherwise. The overall verdict is "not bankable" if any gate is at critical gap, "not yet bankable" if any is at development gap, "bankable subject to conditions" if any is conditional, and "ready for financial close" only if every gate is ready.

Lumora

Lumora's base case is not yet bankable. Four gates sit at development gap: transmission, because the line has no financing; the offtaker, because the collection rate and payment security are below threshold; regulation, because four items are at gap, including tariff pass-through and foreign exchange convertibility; and public finance, because of the contingent exposure, cash needs and negative consolidated fiscal NPV described in Chapters 13 and 14. Financing is ready. The other four gates are conditional.

Table 16.2. Gate status across selected scenarios

Gate	Base	Overrun	Delay	Offtaker	FX step	Transmission	Combined
1. Resource	C	C	C	C	C	C	C
2. Demand	C	C	C	C	C	C	C
3. Configuration	C	C	C	C	C	C	C
4. Transmission	D	D	D	D	D	X	D
5. Offtaker	D	D	D	X	X	D	X
6. Regulation and PPA	D	D	D	D	D	D	D
7. Financing	R	X	D	R	R	R	X
8. Public finance	D	X	D	X	D	D	X
9. Environment and society	C	C	C	C	C	C	C

Note: R = ready, C = conditional, D = development gap, X = critical gap. Debt fixed at base-case amount.

CHAPTER 17

The Lumora case in full

Scenario results

Table 17.1 brings the full-engine runs together. All stress cases hold the commercial debt at its base-case amount and repay it as an annuity, so the base row with locked debt is the right comparator.

Table 17.1. Lumora Falls, PPP structure: scenario results

Scenario	Project IRR	Equity IRR	Min DSC R	Utility capacity ratio	Lifetime utility gap, USD m	Fiscal NPV, central, USD m	Fiscal NPV, consolidated, USD m	Fiscal screen
Base, debt sized	9.7%	16.3%	1.64x	2.11x	0	49	-109	Moderate
Base, debt locked	9.7%	16.3%	1.64x	2.11x	0	49	-109	Moderate
Low case	8.7%	12.4%	1.60x	1.47x	0	-73	-1,156	Moderate
High case	10.4%	18.6%	1.80x	2.12x	0	107	-20	Moderate
Drought	9.5%	15.5%	1.44x	2.11x	0	35	-183	Moderate
CAPEX overrun 27%	7.6%	9.3%	1.61x	2.11x	0	-133	-292	High
CAPEX overrun 96%	4.3%	3.8%	1.55x	2.11x	0	-397	-556	High
Two-year delay	8.3%	11.1%	1.67x	2.33x	0	-54	-184	Moderate
Low demand	9.8%	16.7%	1.67x	1.49x	0	44	-433	Moderate
Offtaker stress	9.7%	16.3%	1.64x	0.00x	5,460	-1,177	-563	High
Offtaker stress, no backstop	n/a	-100.0%	-0.69x	0.00x	5,460	-674	-1,180	High
FX step 50%	9.7%	16.3%	1.64x	0.57x	405	-152	-496	High
Interest +300 bp	9.7%	15.3%	1.54x	2.11x	0	31	-128	Moderate
Transmission 2 yrs late	9.8%	16.5%	1.65x	1.55x	0	47	-227	Moderate

Scenario	Project IRR	Equity IRR	Min DSC R	Utility capacity ratio	Lifetime utility gap, USD m	Fiscal NPV, central, USD m	Fiscal NPV, consolidated, USD m	Fiscal screen
Climate trend	9.6%	16.1%	1.63x	2.11x	0	38	-159	Moderate
Combined	6.4%	7.0%	1.64x	0.00x	5,771	-1,343	-924	High

Note: Combined = CAPEX overrun (27%), construction delay, offtaker stress and FX step together. Utility capacity ratio is the worst of the first ten operating years. Fiscal NPV at 8 percent; consolidated includes the state utility. 'n/a' = IRR undefined because equity is wiped out.

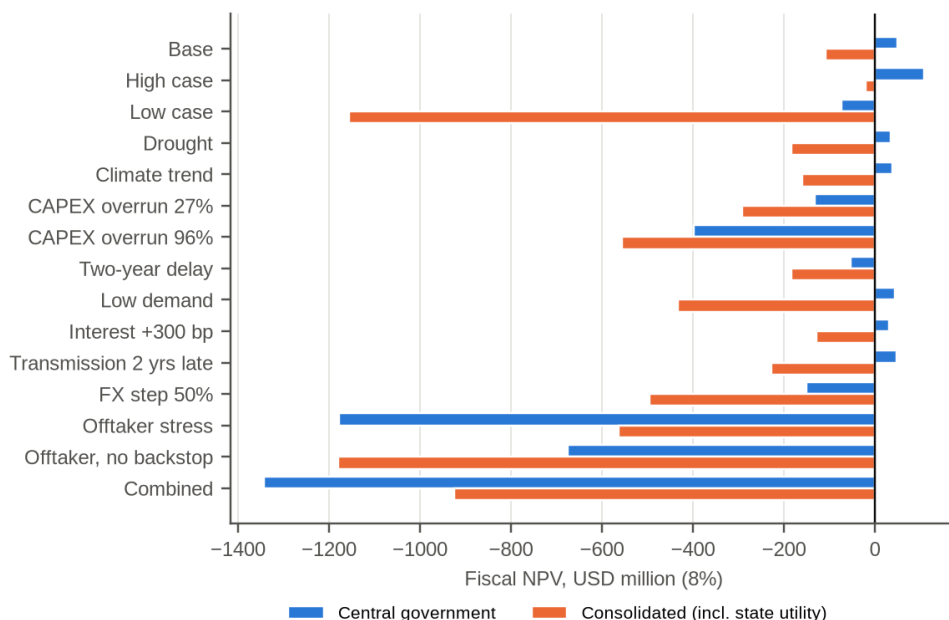


Figure 17.1. Fiscal NPV to the government by scenario, Lumora PPP structure (USD million, 8 percent discount rate). Source: companion model, full-engine runs.

Reading the table

Three patterns stand out.

The project's own metrics look robust. With the budget backstop in place, minimum DSCR stays above 1.4 in every scenario and the project never misses debt service, because the two-part tariff, deemed energy and the backstop protect it. A lender looking only at the project would see a resilient credit. That resilience is borrowed. Remove the backstop under the offtaker stress and the same project defaults.

The utility and the state are not robust. The stresses that matter most for them, offtaker weakness and currency, leave the project untouched and move hundreds of millions of dollars of present value onto the public balance sheet. Even in the base case, the consolidated public sector is worse off with the project at this tariff than without it.

Construction risk is shared. Overrun and delay hurt the sponsors through a lower equity return and a financing gap, and the state through larger public contributions and higher contingent exposure. At the reference-class mean overrun, the gap is too large for any sponsor to fill.

The conclusion is the thesis of the book in numerical form. A project can be bankable on its own terms while the arrangements that make it bankable leave the state carrying risks it has not priced.

CHAPTER 18

Red-team review and limitations

Why we attacked our own model

A model that supports a policy argument should be tested by someone who wants it to fail. Before publication we commissioned an adversarial review of the companion workbook, written from the position of a development-bank model auditor and an examiner in energy economics. The reviewer re-ran the model on separate copies, traced formulas cell by cell and reported thirty findings: three critical, ten high, ten medium and seven low. The workbook passed every one of its own integrity checks while several of these defects were present. That alone justified the exercise.

This chapter reports what the review found, what we changed, and what we did not change. All results elsewhere in this book come from the corrected model.

Critical findings

An uncapped guarantee. The first version let a sovereign PPA payment guarantee pay every dollar of the utility's shortfall, every year, for thirty years. Under the offtaker stress the state ended up paying about 98 percent of all the utility's PPA bills, and the scenario "no budget backstop, guarantee called" produced exactly the same numbers as the scenario with a backstop. Real payment guarantees are capped, usually at a number of months of billing, and are reimbursed when the utility recovers. The corrected model limits calls to the available guarantee amount, tracks calls outstanding and credits the utility's reimbursements back to the Treasury. A shortfall beyond the cap now becomes arrears to the project. The consequence is stark: without a budget backstop, the offtaker stress drives the IPP into default, with a minimum DSCR of $-0.69x$.

Supply that did not exist. The utility model billed the whole of national demand and bought whatever it needed from "other supply" at a fixed cost, with no volume limit, even though the demand sheet showed that committed supply covered only part of that demand. By the end of the horizon the utility was buying several times more energy than existed. This flattered its payment capacity. The corrected model caps other supply at the amount available, so demand above it goes unserved, and treats the mining load served directly by the project as outside the utility's sales.

A fiscal perimeter that stopped at the Treasury. The first fiscal NPV counted only central-government cash. Losses that the state-owned utility absorbed, such as deemed-energy payments for power it could not receive or the extra cost of a PPA priced above its retail realisation, were invisible unless they surfaced as a payment gap. This is why a transmission delay appeared to improve public finances. The corrected model adds a no-project counterfactual for the utility and reports a consolidated fiscal NPV for

government and utility together. The difference is material. In the base case, central-government fiscal NPV is about USD 49 million while the consolidated figure is about USD –109 million, because each megawatt-hour the utility buys from Lumora costs it more than it collects when it resells it.

High-severity findings

The review found that the debt service reserve was never drawn. Shortfalls went straight to guarantors and sponsors while the reserve sat idle, and reserve top-ups were themselves paid out of guarantee calls. The corrected waterfall draws the reserve first, releases only the excess over target, and tops up only from available cash.

"Locked" debt was not locked. In stress runs the first version switched the commercial loan from a sculpted to an annuity profile and let the concessional loan, grants and government equity grow with cost. Stress results therefore mixed the effect of the stress with the effect of refinancing. The corrected model locks every tranche amount and the commercial principal schedule at their base-case values, lets interest float only on the unhedged share, and lets equity absorb the difference, which is what a sponsor faces after financial close.

The low-demand stress did not propagate, because the base-year demand used to scale system peak and utility costs was itself scaled by the stress. A transmission delay was modelled as a rescheduled line, which made it cheaper in present-value terms; it is now a late line with a cost penalty, built to the original plan. Project-funded transmission spent after commercial operation dropped out of the funding and cash-flow accounts; it now reduces cash flow available for debt service, and a new check reconciles it.

The original financing gap was largely an artefact. Interest during construction, fees and the reserve were funded by equity while the gearing limit applied only to capital cost, so effective gearing on total uses was about 65 percent rather than 75. The corrected model funds IDC and fees with debt through a closed-form factor that avoids circularity, and the gap in the base case disappears.

Gate metrics could be text or blank. A failed internal rate of return calculation scored as ready, and a utility with no exposure scored as a critical gap. Both are fixed with explicit fallbacks.

The closed-form structure comparison omitted IDC and tax and so understated the tariff an IPP needs. The corrected screen grosses equity returns up for tax and includes IDC, and the book now reports tariffs solved by the full engine alongside it. The reviewer also pointed out that the result "gross public exposure barely changes across structures" holds by construction, because every dollar of capital is either public or protected by a termination payment. We kept the result but now present it as an identity, which is how it should be read.

Contingent exposures mixed stocks and flows, and the termination amount collapsed to the outstanding debt once equity had been repaid. The corrected model treats foreign

exchange cover inside the maximum, values equity at termination as the larger of unrecovered equity with a premium and the present value of remaining distributions, and lets the state recover debt-guarantee calls from later project cash ahead of dividends.

Finally, the stress magnitudes did not match the model's own evidence base. The source database recommends the reference-class median for an expected outcome and the mean for a stress, but the model used the median as its stress. We kept the median as the default overrun stress and added a run at the reference-class mean of 96 percent. We also deepened the drought stress to 55 percent of normal flow, in line with the cut in Kariba's 2024 water allocation.

Medium and low findings

We fixed several smaller items. The LLCR now covers every tranche over the longest loan life at the weighted interest rate. One regulatory item at critical gap now makes Gate 6 critical. Gate 8 now tests on-budget debt and the consolidated fiscal NPV. Tax losses are no longer consumed during the tax holiday. Royalties are charged only on energy delivered. The on-budget screening measure adds components in the same year rather than adding peaks from different years. Three new integrity checks cover the debt balloon, the reserve balance and project-funded transmission.

What we did not change

Some limitations remain, and readers should weigh the results with them in mind.

- The lender case still applies a one-year P90 to every year. A ten-year P90 would be closer to P50, so the lender case is conservative on average and silent on runs of dry years. Debt sizing is set by the gearing cap rather than by coverage in most structures, so this matters less than it might.
- Annual periodicity and proportional curtailment remain. A plant with storage would do better than the model suggests when the line is constrained.
- The currency step is permanent in real terms, with no pass-through to local prices. This is severe but not unprecedented.
- Debt is drawn in proportion to spending rather than after equity, which slightly flatters the equity return.
- The utility model has no balance sheet and does not revalue receivables for currency moves.
- Call probabilities for expected loss remain user judgements.
- Only one project is assessed. The portfolio question raised in Chapter 19 is not modelled.

The review also flagged claims in the earlier draft that the model did not support, including figures for debt capacity, the LLCR and the effect of a transmission delay on

public finances. Those passages have been rewritten, and every figure in this edition is generated from the corrected model.

CHAPTER 19

Rules for financial close

The rules below are our synthesis of the evidence and the modelling in this book. They are written for the people who negotiate and approve hydro transactions, and they are deliberately blunt.

1. Price and size on P90, and also test a run of dry years. A single-year statistic says nothing about persistence, and persistence is what breaks debt service.
2. Budget against the reference class. A median real overrun of 27 percent is an expectation, not a stress [17].
3. Finance the line with the plant. Same close, same timetable, and an interface agreement that says who pays deemed energy and for how long.
4. Compute the buyer's maximum sustainable payment before negotiating the tariff. If the bill exceeds it, decide openly who will fund the difference.
5. Match currencies where you can. Local-currency tranches, partial local-currency tariffs or explicit sharing of devaluation risk are cheaper than a guarantee called after a crisis.
6. Treat every guarantee as debt that has not yet arrived. Record it, cap it and disclose it, as the PPP disclosure framework and the IMF fiscal transparency code recommend [49, 50].
7. Tie every grant to a lower tariff or a cap on returns. Otherwise public money becomes private profit.
8. Look at the portfolio, not only the project. One PPA may pass the screen while the sum of all of them does not. The Lao experience is the warning [4].
9. Procure competitively and in the open. It decides who will lend, and on what terms.
10. Measure the fiscal effect on the whole public sector, utility included. A Treasury-only view can show a gain where the state as a whole loses.
11. Close only when no gate is at critical gap and the Ministry of Finance has signed off the fiscal screen in writing.

ANNEXES

Reference material

ANNEX A

Case library summary

The full case files cover twenty-one African and seven international projects across thirty fields each, and record "public data not found" wherever a figure could not be verified. Table A.1 summarises the cases with a cost figure in US dollars. Unit costs are nominal, unadjusted for price level and inconsistent in scope; they are indicative only.

Table A.1. Public benchmark cases (as published; nominal, unadjusted)

Project	Country	Structure	M W	USD/kW	Timeline	Credit support
Nachtigal	Cameroon	IPP/PPP, IFC-led DFI debt	42 0	3,293	FC Nov 2018; plan 5 yrs; full COD slipped to Dec 2024 / 2025	IBRD payment guarantee EUR 86m + loan guarantee EUR 171m; MIGA; 75:25 debt:equity
Bujagali	Uganda	IPP + IDA PRG	25 0	3,448	c.2007-2012	IDA PRG US\$115m with sovereign indemnity; MIGA US\$115m; liquidity facility
Bui	Ghana	Public (BPA), China Exim, cocoa escrow	40 0	1,975	COD Dec 2013	China Exim tranches secured on cocoa export escrow
Kafue Gorge Lower	Zambia	State-owned, sovereign- guaranteed Chinese debt	75 0	2,667	Nov 2015 → full COD Mar 2023	Sovereign guarantee; ZESCO arrears entered sovereign DSA
Isimba	Uganda	Public, 85% China Exim / 15% GoU	18 3	3,102	Apr 2015 → Mar 2019	Sovereign borrower
Karuma	Uganda	Public, 85% China Exim / 15% GoU	60 0	2,833	Contract 2013 (5 yrs) → COD Jun 2024	Sovereign borrower
Julius Nyerere (Rufiji)	Tanzania	100% state budget	2,1 15	1,348	Construction 2019 → 2025/26 (42-month EPC)	None (budget-funded)
Rusumo Falls	Rwanda/ Tanzania/ Burundi	Regional public, IDA-financed	80	5,861	Approval 2013 → COD Feb 2025	IDA grants/credits; transmission parallel- financed AfDB/EU

Project	Country	Structure	M W	USD/kW	Timeline	Credit support
Mpatamanga	Malawi	PPP, storage + peaking	35 8	4,563	Pre-construction; 5-yr build in 35-yr PPA	IDA PRG US\$100m; MIGA US\$180m; IDA grant US\$350m to govt
Ruzizi III	DRC/ Rwanda/ Burundi	Regional PPP (stalled)	20 6	3,689	Target 2030 before conflict pause	IFIs ~60% expected; EIB lead arranger
GERD	Ethiopia	State, domestically financed	5,1 50	971	2011 → inauguration Sep 2025	Domestic bonds/budget; no MDB finance
Gibe III	Ethiopia	State, ICBC- financed EPC	1,8 70	n/a	Inaugurated Dec 2016	ICBC loan after MDBs withdrew
Nam Theun 2 (intl.)	Lao PDR	Export IPP, MDB-guaranteed	1,0 70	1,355	FC Jun 2005 → COD Apr 2010 (snippet)	IDA & ADB PRGs, MIGA (~US\$186m debt covered); EGAT take-or-pay PPA
Upper Trishuli-1 (intl.)	Nepal	DFI-led domestic IPP	21 6	2,997	Completion slipped 2024 → Dec 2026 (snippet)	DFI debt ~70%
Reventazón (intl.)	Costa Rica	Utility trust & lease, project bond	30 6	4,583	FC Jan 2014 → inaugurated Sep 2016 (snippet)	Trust owns plant, leased to ICE; IDB A- loan, IFC, IG-rated bond, local-currency bank debt

Note: Sources for each row are listed in the case files; costs in EUR are not converted (n/a). 'Snippet' marks a fact seen only in search results.

ANNEX B

Benchmark ranges

Table B.1. Benchmark ranges used to set or check model defaults

Parameter	Low	Central	High	Unit	Source
Hydro CAPEX - large (>10 MW), Africa	2,330	2,515	3,290	2024 USD/kW	[11] [15] [51]
Hydro CAPEX - small (<=10 MW), Africa	3,795	4,084	4,200	2024 USD/kW	[11] [51]
Hydro CAPEX - global weighted average (reference)	1,494	2,267	3,126	2024 USD/kW	[11]
Capacity factor - large hydro, Africa	33%	48%	81%	% of hours	[11]
Capacity factor - small hydro, Africa	50%	55%	66%	% of hours	[11]
Fixed O&M	1.0%	2.0%	3.0%	% of total installed cost per year	[11]
LCOE - hydro, global weighted average 2024	-	0.057	-	2024 USD/kWh	[11] [15]
WACC assumption - Africa	-	12%	-	% (real, IRENA assumption)	[11] [52]
Construction cost overrun (real, vs. budget at approval)	27%	-	96%	% of approved budget	[17]
Schedule overrun	27%	-	44%	% of planned implementation period	[17]
Implementation / build duration - large hydro	7	8.6	10	years	[17] [11] [53]
Funded EPC contingency	-	20%	-	% of EPC contract value	[15]
Senior debt tenor (hard currency)	15	18	21	years	[35] [15]
Gearing (debt : equity)	-	76:24	-	ratio	[15]
DSCR - lenders' base case	1.45x	1.53x	-	x	[15]
Construction-period grace	-	~6	-	years	[15]
Utility T&D losses, SSA	10%	15%	23%	% of energy dispatched	[5]
Utility bill collection rate, SSA	58%	-	100%	% of billed	[5]
Power-sector quasi-fiscal deficit	-	1.5%	>5%	% of GDP	[5] [25]
Fiscal injections to power SOEs	-	0.25%	>0.6%	% of GDP per year	[6]

Parameter	Low	Central	High	Unit	Source
Power PPP renegotiation incidence	24%	-	41%	% of contracts	[6]
Fiscal cost of contingent-liability realisation	1.2%	3%	6%	% of GDP per event	[39]
LIC-DSF contingent-liability stress shock	2%	35%	5%	see notes	[42]
LIC-DSF PV of PPG external debt / GDP threshold	30%	40%	55%	% of GDP	[42]
Transmission line cost, 400 kV AC (East Africa)	0.36	0.42	0.70	M USD(2012)/km incl. IDC	[23]
Transmission line cost, 220 kV AC (East Africa)	0.27	0.36	0.63	M USD(2012)/km incl. IDC	[23]
Transmission line cost, 330/400 kV interconnectors (SAPP)	0.19	0.23	0.38	M USD/km (2013 study)	[54]
Transmission line cost, HVDC 500-600 kV	0.61	0.94	1.41	M USD(2012)/km incl. IDC	[23]
Interconnector / transmission-level losses	1%	2-3%	7%	% of energy transmitted	[54]
Liquidity facility size (RLSF)	-	12	-	months of revenue (max)	[48]
Drought: reduction in hydro water allocation (Kariba 2024)	-	47%	~85%	% reduction vs normal	[12] [55]

Note: Full notes and confidence ratings for each benchmark are in the research database.

ANNEX C

Model map

Table C.1. Sheets of the companion model

Sheets	Content
00, 01	Read-me and control panel: case, generation case, structure, debt mode, backstop, nine stress toggles, five flexes
02 to 07	Project inputs, hydrology, generation, plant cost, construction timeline, operating cost
08 to 10	Transmission, grid, demand
11, 12	Offtaker and utility cash model
13 to 16	Regulatory matrix, PPA, tariff, revenue and shortfall routing
17, 17A, 18 to 21	Sources and uses, five structures, debt, equity, cash flow, tax
22 to 26	Government support, guarantees, contingent liabilities, fiscal impact, fiscal screening
27, 28	Scenarios and sensitivities with full-engine snapshots
29, 30	Risk allocation and the nine-gate framework
31 to 33	Case benchmarks, dashboard, integrity checks

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